2024

COMPUTER SCIENCE — HONOURS

Paper : DSCC-3

(Data Structure)

Full Marks : 75

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

1. Answer ***any five*** questions : 2×5

   (a) What are sparse matrices?

   (b) Why is queue called FIFO?

   (c) Differentiate between linear and non-linear data structure.

   (d) Write the advantages of circular queue over linear queue.

   (e) What is the prerequisite for binary search?

   (f) State the characteristics of a good hash function.

   (g) Define max and min heap.

   (h) Evaluate the following post-fix expression :

   12 7 3 – / 2 1 5 + * –

Section – A

Answer ***any three*** of the following questions.

2. Write down the algorithm of Quick Sort. 5

3. (a) Create a binary search tree with the following values (Show each step) :

   8, 10, 3, 9, 7, 15, 14, 6, 2, 3

   (b) What are the advantages of threaded binary tree? 3+2

4. What is Collision? How can we resolve it? 2+3

5. Polynomials can be represented either by an array or by linked list. Compare and contrast these two types of representations. Represent the following polynomial by a linked data structure (show only the diagram) : 3+2

$$-5x^5 + 4x^4 - 25x^3 + 10.$$

6. Write the difference between recursion and iteration. State the advantages and disadvantages of both the types. 3+2

Section – B

Answer ***any five*** of the following questions.

7. Write an algorithm to convert an infix expression to its postfix form using stack.
Convert : A+(B*C–(D/E^F)*G)*H into its postfix form. 5+5

8. (a) Draw the binary tree with the following orders given below :
Pre-Order : A, B, D, I, E, J, C, F, G, K and In-Order : D, I, B, E, J, A, F, C, K, G.
Show all the steps.

(b) Mention the advantages of binary search over linear search.

(c) Find the time complexity of binary search method. 4+2+4

9. (a) Write a C function to print of a linear linked list in reverse order and also count the number of nodes of a linear linked list.

(b) In which data structure insertion and deletion of elements can take place from either end? Explain. 5+5

10. (a) Write the algorithm of Merge Sort.

(b) Compare Insertion sort, Radix sort and Shell sort with respect to their time complexity.

(c) Is Merge sort an example of in-place sort? Justify your answer. 5+3+2

11. (a) Write down Heap sort algorithm.

(b) Sort the following values using Heap sort (Show each step) :
12, 3, 9, 14, 10, 18, 8, 23 5+5

12. (a) Insert following key values in a Hash table of size 11. Use Module division method and Linear Probing.
55, 32, 15, 37, 48, 68, 11, 50, 70, 81.

(b) Discuss Double Hashing and Chaining with suitable examples. 5+5

13. (a) Consider the array *a* [10] [10] whose base address is 3000. Calculate the address of the location *a* [2] [3] using both row and column major orderings. Assume that the index of the array starts at *a* [1] [1] and the array stores integers of size 4 bytes each.

(b) Write an algorithm to insert a node X to a double linked list. Consider all cases. 4+6
